Hybrid Quantum–Classical Campaign Optimization

A leakage-controlled study of response prediction, constrained customer selection, and quantum-circuit evaluation

Final challenge answer
Selected six: C0028 · C0031 · C0079 · C0081 · C0327 · C0338
Exact independent-response reference: **P(K ≥ 3) = 0.2078515973394121**

0.2417	16.1710	0.207852	31 s
Stage 1 mean PR-AUC	Stage 2 exact objective	Stage 3 exact probability	IBM QPU time recorded

Three stages, one connected pipeline

A classical model ranks customers using training data. An exact classical oracle selects six from the frozen top ten. A six-qubit circuit encodes their independent success probabilities. Separately, a synthetic QAOA instance was sampled on IBM hardware to compare circuit depths. This is an empirical implementation study; it does not demonstrate quantum advantage.

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1 / Response prediction

Data and protocol

The model comparison used 300 labeled Stage1_Train rows (35 positive, 265 negative), with three stratified folds repeated three times. Two-fold inner selection used average precision. Imputation, encoding, scaling, model fitting, and positive-slope sigmoid calibration were fit within each training partition. Customer ID and duration were excluded as predictors; Stage1_Test labels were not loaded or used.

CatBoost was the provisional reference by mean outer-fold average precision. Fold standard deviations are descriptive: repeated folds overlap and are not independent confidence intervals.

CatBoost validation metric	Raw	Calibrated
ROC-AUC	0.6555	0.6555
Average precision	0.2417	0.2417
Precision@10 / Recall@10	0.189 / 0.162	0.189 / 0.162
Brier score / Log loss	0.1523 / 0.4767	0.1012 / 0.3526
10-bin ECE	0.1833	0.0183

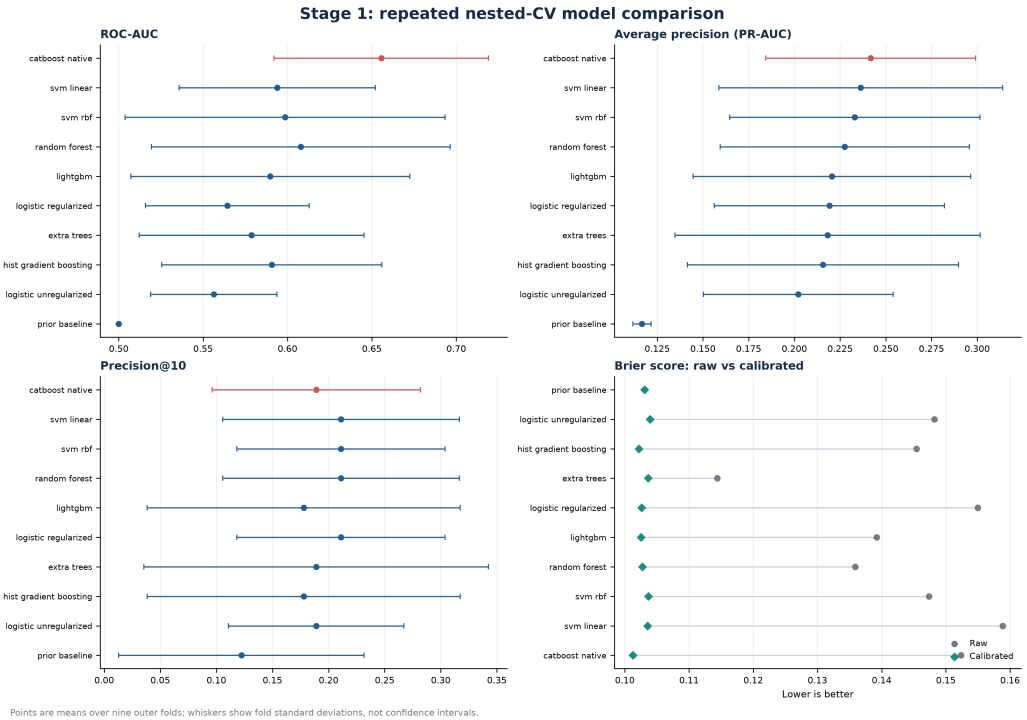


Figure 1. Ten classical candidates across nine outer folds. Whiskers show fold standard deviation, not confidence intervals.

Stage 1 / Ranking and reliability

Average precision emphasizes the minority positive class; ROC-AUC provides a prevalence-insensitive ranking view. These curves aggregate repeated out-of-fold predictions and should be interpreted alongside fold-level variation.

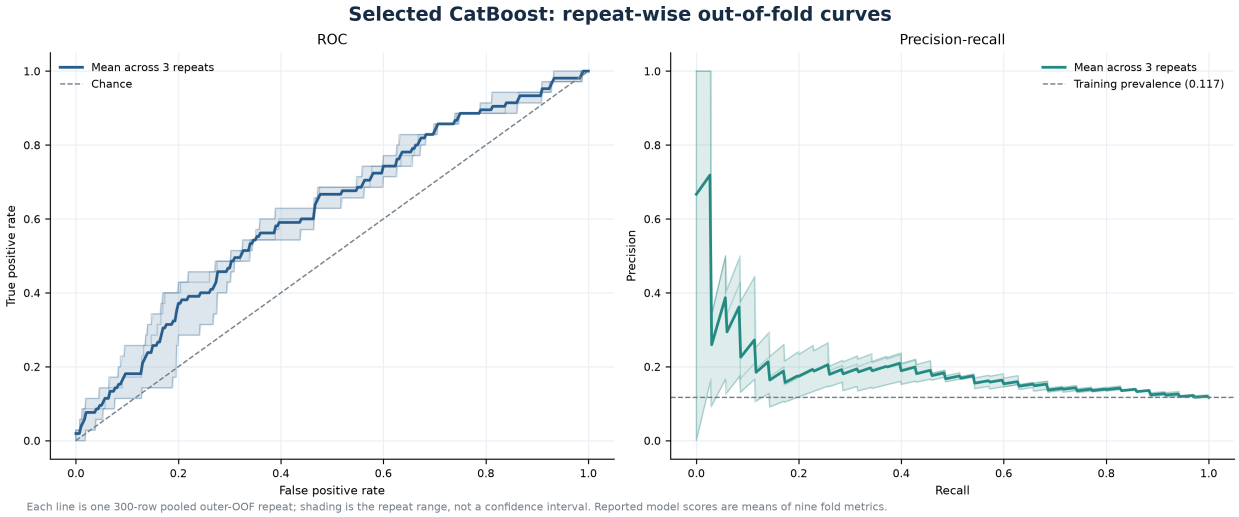


Figure 2. CatBoost ROC and precision-recall performance across repeated out-of-fold predictions.

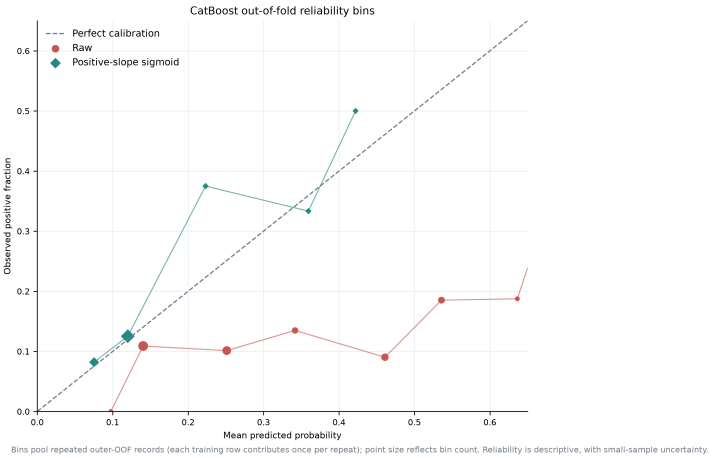


Figure 3. Raw and calibrated reliability bins. Calibration reduces Brier score from 0.1523 to 0.1012 and log loss from 0.4767 to 0.3526; generalization under distribution shift is not established.

2 / Exact customer selection

The frozen inference probabilities defined the ten-customer candidate pool. All 210 six-customer subsets were enumerated and cross-checked with an independently encoded SciPy MILP. Both methods select the same portfolio. All 210 subsets in this pool satisfy budget and capacity, so those constraints are nonbinding for the final shortlist.

Exact objective: 16.1710314161
Budget: 16 / 25 contact-cost units **Capacity:** 15 / 22 units
Enumeration–MILP gap: 0

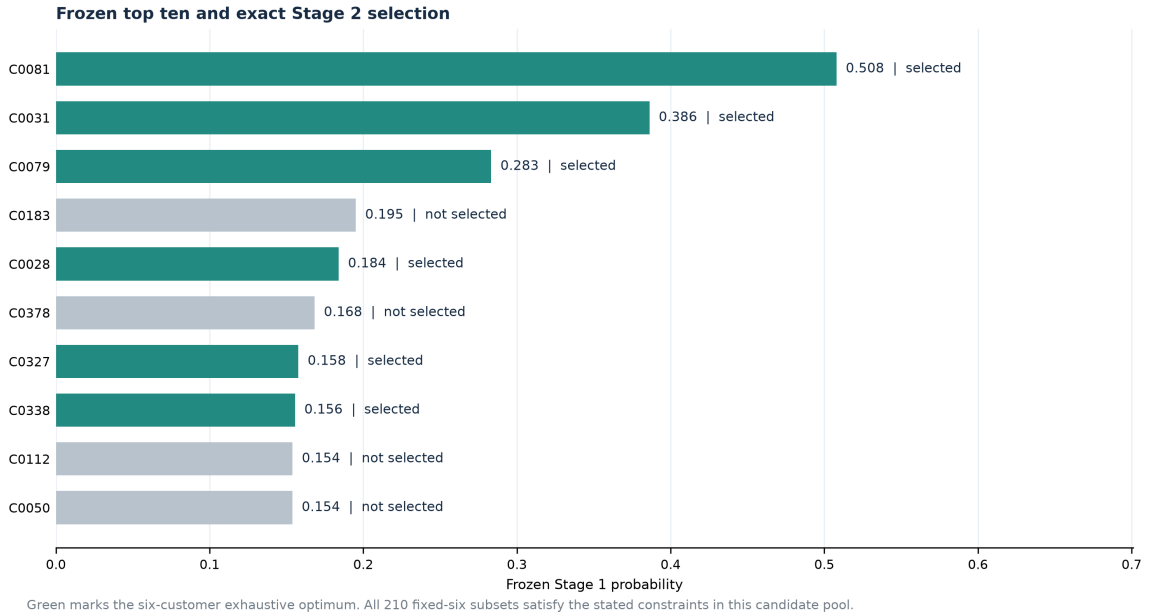


Figure 4. Frozen top ten ranked by Stage 1 probability; highlighted customers are the exact six selected using the Stage 2 objective.

Selected portfolio

Customer	p(response)	Expected value	Contact cost	Capacity
C0028	0.184046	27	4	4
C0031	0.386291	13	2	1
C0079	0.283243	24	3	3
C0081	0.507980	17	4	1
C0327	0.157800	25	1	4
C0338	0.155640	18	2	2

The objective is exact conditional on the frozen Stage 1 probabilities and the supplied operational values. It does not remove uncertainty in the predicted probabilities.

3 / QAOA diagnostic on the actual shortlist

The actual ten-candidate pool was also evaluated with an exact-six XY-QAOA circuit. The additive cost phase is exact on this fixed-six sector because all 210 subsets are feasible. Four single-seed fits compare COBYLA/SPSA and expectation/CVaR objectives; each was capped at ten evaluations. None is convergence-certified.

> This is a real Qiskit circuit and Aer diagnostic, but parameter optimization was severely budget-limited. It is not the later IBM hardware QAOA experiment, which uses a separate synthetic instance.

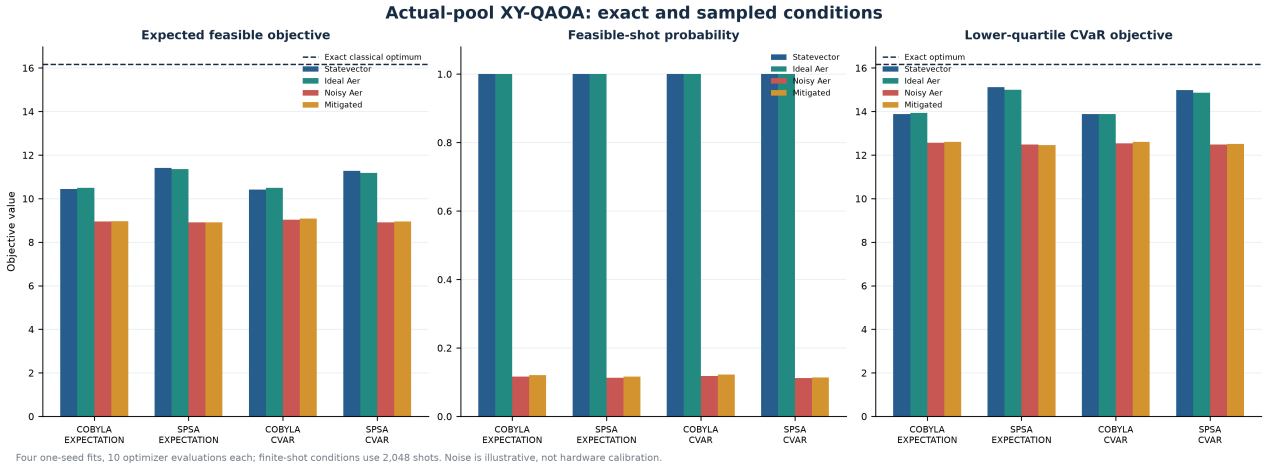


Figure 5. Actual-pool QAOA objective and feasibility comparisons from saved statevector and Aer results. Exact classical optimization is the reference.

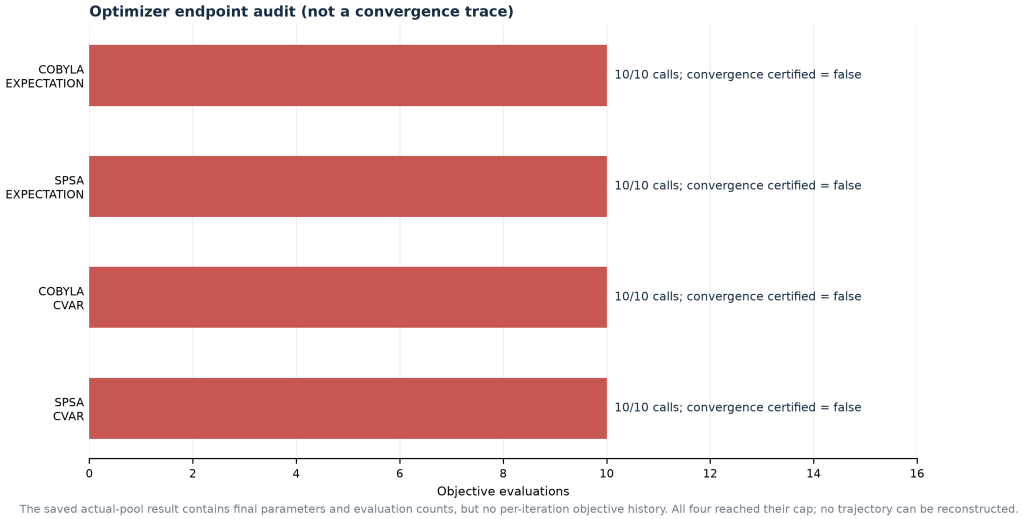


Figure 6. Saved optimizer evaluation counts and termination status. No convergence trajectory was saved or reconstructed.

The exact optimum objective is 16.1710. The small-budget QAOA endpoints remain below that reference; sampled circuits are diagnostic and do not replace exact selection.

4 / Stage 3 probability circuit

Assuming independent customer responses X■ ~ Bernoulli(p■), the exact target is P(K ≥ 3), where K = ΣX■. A Poisson-binomial dynamic program and an explicit enumeration of all 64 bitstrings agree at **0.2078515973394121**. The ideal Qiskit statevector agrees to numerical precision.

For each customer, the circuit prepares RY(2 asin √p■)|0■ and measures q■ into c■. Qiskit prints bitstrings as c■…c■; therefore q■ is at character position 5–i. The tests explicitly validate this mapping.

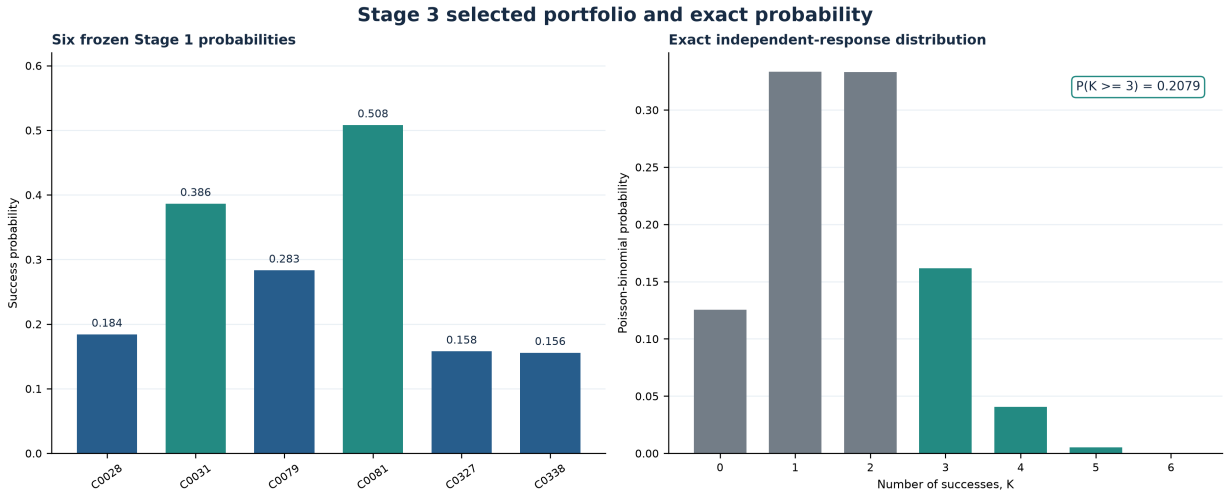


Figure 7. The six frozen Stage 1 probabilities and their exact Poisson-binomial distribution; the target event is K ≥ 3.

Input and individual-qubit validation

Qubit	Customer	p■	RY angle	Statevector p■
0	C0028	0.184046	0.886783	0.184046
1	C0031	0.386291	1.341371	0.386291
2	C0079	0.283243	1.122407	0.283243
3	C0081	0.507980	1.586757	0.507980
4	C0327	0.157800	0.817016	0.157800
5	C0338	0.155640	0.811075	0.155640

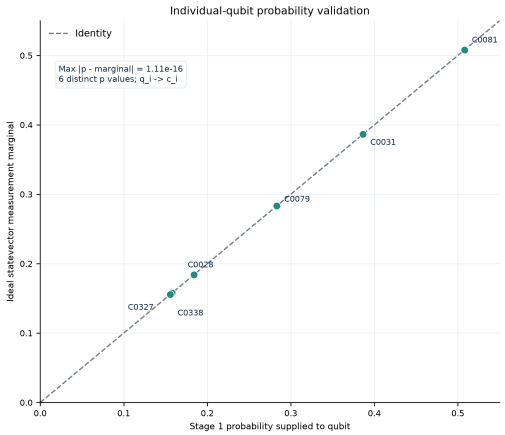


Figure 8. Statevector marginals match their customer-specific target probabilities.

This six-qubit product-state circuit has no entangling gates. It validates probability preparation and bit ordering; it does not model dependence between customer outcomes.

5 / Sampling, noise, and Stage 3 hardware

Local simulation covered 512, 2,048, 8,192, and 32,768 shots over three seeds. Readout mitigation uses estimated assignment matrices and a projected quasi-distribution; its bootstrap intervals condition on the calibration estimate. Mitigation did not consistently reduce error and increased estimated variance.

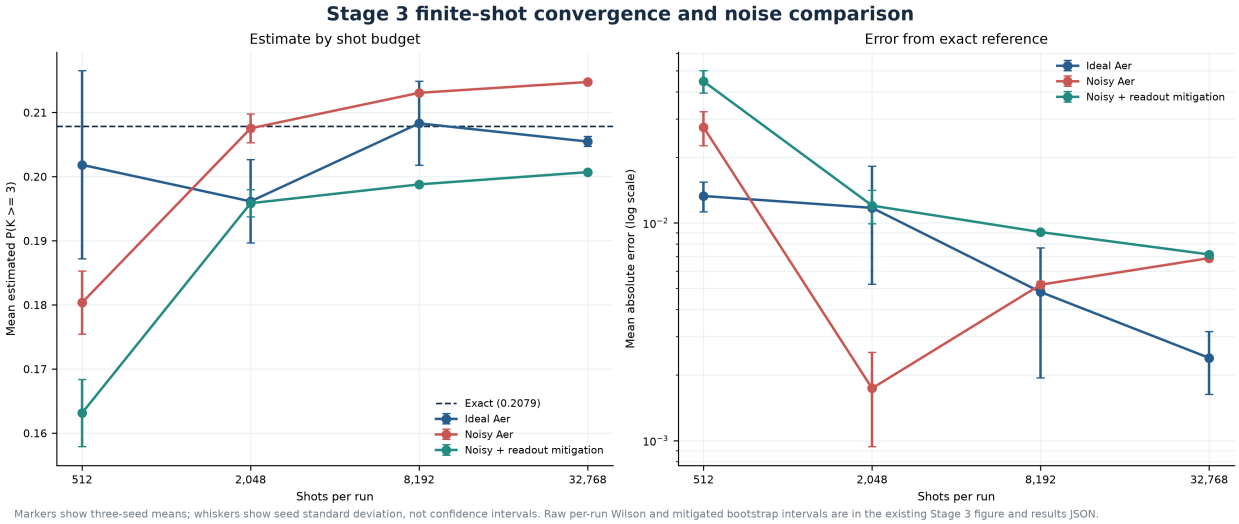


Figure 9. Saved Stage 3 estimates across shot budgets. Whiskers show seed variation, not confidence intervals.

A later Colab run executed the same six-qubit Stage 3 circuit on IBM hardware. This is actual physical QPU execution, separate from the local Aer studies.

Stage 3 condition	Estimate	95% interval	Shots
Exact Poisson-binomial	0.207852	Exact	—
IBM Marrakesh, raw	0.201538	0.192992–0.210364	8,192
IBM Marrakesh, readout mitigated	0.202275	0.192653–0.211392	8,192

Job dbqfi99abdes73fo6vig • backend ibm_marrakesh • qubits [87, 6, 10, 139, 120, 107] • 5.0 QPU seconds. The paired mitigation effect is inconclusive; its uncertainty interval includes zero.

The source export's resumed registry incorrectly assigned Kingston metadata to this earlier Marrakesh job. The job ID and counts are unchanged. Canonical records use the original submission backend/layout; the immutable export and full correction rationale are retained in the repository.

6 / Synthetic QAOA hardware comparison

One IBM job evaluated fixed, noise-free-optimized exact-cardinality XY-QAOA parameters at p=1 and p=2. Each depth received 30,000 hardware shots on the same six-qubit chain, plus readout calibration circuits. The objective is a synthetic six-candidate choose-three instance, not the bank-customer selection. There was no hardware-in-loop optimization.

Depth / condition	Feasibility	P(exact optimum)	Approx. ratio	95% ratio interval
p=1 Hardware raw	0.815	0.0164	0.4706	0.4681–0.4732
p=1 Readout mitigated	0.916	0.0171	0.4729	0.4708–0.4753
p=2 Hardware raw	0.761	0.0056	0.6240	0.6224–0.6256
p=2 Readout mitigated	0.882	0.0054	0.6245	0.6228–0.6261

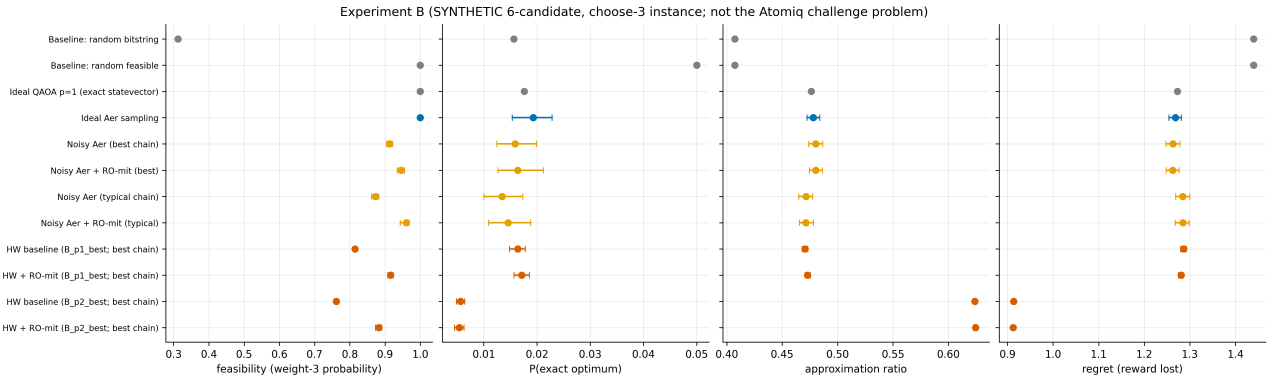


Figure 9. Hardware and simulation quality for the synthetic QAOA instance. Values do not describe customer selection.

The p=2 approximation ratio is higher than p=1, while its exact-optimum sampling probability is lower. Readout correction increases measured feasibility; ratio changes are small and intervals overlap. The exact classical optimum remains authoritative. No quantum advantage or general mitigation gain is established.

> Job dbqg5d1abdes73fo7o10 • ibm_kingston • qubits [147, 148, 149, 150, 151, 152] • 30,000 shots per depth • 26.0 QPU seconds. Combined usage across the two hardware jobs: 31.0 seconds.

7 / Reproduction and limitations

What these results support

The Stage 2 result is an exact optimum conditional on the frozen top ten, probabilities, objective, and constraints. The Stage 3 exact probability is valid under independent Bernoulli responses. The IBM records show that the declared circuits were submitted and returned hardware counts. The comparisons do not show a quantum speedup or quantum advantage.

Known limits

Stage 1 has only 300 labeled rows and 35 positive cases; repeated-fold estimates and calibration are noisy. All final-pool Stage 2 subsets are feasible, so the challenge instance does not test binding constraints. The actual-pool QAOA optimizer stopped at ten evaluations and is not convergence-certified. The hardware QAOA objective is synthetic. Stage 3 inherits uncertainty in Stage 1 probabilities and assumes independent outcomes. The original challenge PDF/workbook are not redistributed because permission is unverified.

Reproducibility

The local locked environment is CPython 3.12 with Qiskit 2.6.2 and Aer 0.18.2. The hardware Colab run recorded Python 3.13.18, Qiskit 2.2.3, Aer 0.17.2, and Runtime 0.43.1 (seed 2026); these are distinct environments. The final local test run passed 61 tests. The public notebook is disarmed by default and contains no execution output or API key. Hardware evidence, raw counts, calibration exports, and provenance correction are under results/ibm_hardware/.

Task	Command / authoritative record
Install	py -3.12 -m venv .venv; install requirements-lock.txt
Tests	.venv/Scripts/python.exe -m pytest -q
Stage 1	scripts/run_phase1_classical.py; see results/stage1_classical/
Final result audit	scripts/audit_actual_results.py; results/final_actual_independent_audit.json
Hardware evidence	reports/ibm_hardware_execution_audit.md; results/ibm_hardware/

Report values are taken from saved artifacts and the passed independent audit. Report generation is deterministic post-processing only; it did not rerun models, optimizers, Aer, or IBM jobs.

Project repository: github.com/Athleity/Atomiq-Hybrid-Quantum-Optimization • Computational baseline cfe9afc • Public sanitized submission commit 4d9c9a5